Structural modification of azolylacryloyl derivatives yields a novel class of covalent modifiers...

Search within citing articles

Red blood cells as therapeutic target to treat sickle cell disease

S Bhatt, DA Argueta, K Gupta… - Antioxidants & redox …, 2024 - journals.sagepub.com
Significance: Sickle cell disease (SCD) is the most common inherited diathesis affecting mostly underserved populations globally. SCD is characterized by chronic pain and fatigue …
☆ Save Cite Cited by 17 Related articles All 4 versions

Modulating hemoglobin allostery for treatment of sickle cell disease: current progress and intellectual property

PP Pagare, A Rastegar, O Abdulmalik… - Expert opinion on …, 2022 - Taylor & Francis
Introduction Sickle cell disease (SCD) is a debilitating inherited disorder that affects millions worldwide. Four novel SCD therapeutics have been approved, including the hemoglobin …
☆ Save Cite Cited by 24 Related articles All 4 versions

Isolation and Characterisation of Quercitrin as a Potent Anti-Sickle Cell Anaemia Agent from *Alchornea cordifolia*

O Adeniyi, R Baptista, S Bhowmick, A Cookson… - Journal of clinical …, 2022 - mdpi.com
Alchornea cordifolia Müll. Arg.(commonly known as Christmas Bush) has been used traditionally in Africa to treat sickle cell anaemia (a recessive disease, arising from the S …
☆ Save Cite Cited by 25 Related articles All 17 versions »

Michael acceptor compounds as hemoglobin oxygen affinity modulators for reversing sickling of red blood cells

KA Mohammad, AH Naghi, MS Ghatge, B Balogun… - Pharmaceuticals, 2025 - mdpi.com
Background/Objectives: Sickle cell disease (SCD) is caused by a β-globin gene mutation (βGlu6Val) that produces sickle hemoglobin (HbS). When deoxygenated, HbS polymerizes …
☆ Save Cite Cited by 1 Related articles All 8 versions »

Targeted modification of furan-2-carboxaldehydes into Michael acceptor analogs yielded long-acting hemoglobin modulators with dual antisickling activities

AM Omar, O Abdulmalik, KM El-Say… - Chemical biology & …, 2024 - Wiley Online Library
Sickle cell disease (SCD) is the most common genetic disorder, affecting millions of people worldwide. Aromatic aldehydes, which increase the oxygen affinity of human hemoglobin to …
☆ Save Cite Cited by 6 Related articles All 7 versions

An investigation of structure-activity relationships of azolylacryloyl derivatives yielded potent and long-acting hemoglobin modulators for reversing erythrocyte …

AM Omar, O Abdulmalik, MS Ghatge, YA Muhammad… - Biomolecules, 2020 - mdpi.com
Aromatic aldehydes that bind to sickle hemoglobin (HbS) to increase the protein oxygen affinity and/or directly inhibit HbS polymer formation to prevent the pathological hypoxia …
☆ Save Cite Cited by 11 Related articles All 12 versions »

Hemoglobin: A Target for Sickle Cell Disease Drug Discovery

AK Donkor, A Opare, W Thomas, O Abdulmalik… - … Approaches to Sickle …, 2026 - Springer
Adult human hemoglobin (herein referred to simply as Hb) plays a crucial role in various physiological functions, with oxygen transport being the primary function. However, Hb is …
☆ Save Cite Related articles All 4 versions

Challenges of the allosteric regulation of hemoglobin by interaction with Cys93 of the β chain for the treatment of sickle cell disease

EJ Niesor, A Perez, C Bordat… - Journal of Sickle Cell …, 2025 - academic.oup.com
Voxelotor exerts its regulatory effect on the affinity of hemoglobin for oxygen through an allosteric mechanism involving binding of the drug to the α-chain of hemoglobin. Cystein 93 …
☆ Save Cite Cited by 1 Related articles All 2 versions

Isolation and Characterisation of Quercitrin, a Potent Anti-sickle Cell Anaemia Agent from the Nigerian Shrub, Alchornea Spp

OA Adeniyi - 2020 - research.aber.ac.uk
Alchornea spp. is an important plant genus in Africa, where it has been used traditionally by healers for the treatment of sickle cell anaemia (SCA). SCA is a genetic disorder endemic to …
☆ Save Cite Related articles All 3 versions »